



AMMUNITION BEFORE 1900

Smoothbore guns and rifles were loaded at the muzzle with lead balls and a separate propellant (gunpowder), ignited by fine gunpowder acting as primer. Guns became easier to load with the advent of the cartridge, a package carrying the lead ball and propellant. While early paper cartridges had to be torn open, later ones could be loaded whole. It was the unitary metallic cartridge (see pp.112–13), a combination of cartridge and primer in one case, that made breech-loading quick and simple.

The powder-and-ball era

To achieve any sort of accuracy, the ball fired from a smoothbore gun had to be spherical and of an exact size. Rifling improved matters, but made the weapon slow to load; the problem was solved by the expanding bullet (see pp.98–99).



MUSKET/RIFLE BALLS

The size of the ball was expressed in “bore,” being the number of balls of that size that could be cast from 1lb (0.45kg) of lead.

BELTED BALL

Some balls, such as the Brunswick ball (see p.98), were belted to slide into the grooves in a gun’s rifled barrel.



MINIÉ BULLETS

These bullets had a hollow base. The force of the propellant detonating caused the bullets’ skirts to expand and grip the rifling.

GROOVED MINIÉ BULLET

Greased grooves in the bullet lubricated the barrel as the bullet gripped the rifling.



PERCUSSION CAPS

The percussion caps (see pp.80–81) provided an easier way to ignite the propellant by using a chemical primer. It was a thin, copper cap shaped to fit over a hollow plug attached to the breech of the gun. The chemical in it exploded when struck by the gun’s hammer. Percussion caps could be used with powder and ball, as well as the earliest cartridges.

Early cartridges

Early 19th-century cartridges carried a measured quantity of gunpowder and a bullet. Wrapped in paper, skin, or fabric, these cartridges posed a problem for breech-loading guns, whose breeches had to be sealed to prevent leakage of gases produced by the ignited propellant. To propel the bullet efficiently, a gas-tight seal was needed at the breech. The solution lay in the metallic cartridge, which was able to seal the breech perfectly. At the same time, metallic cartridges became “unitary” cartridges by integrating the primer, along with the propellant and projectile, within their metal shell. Metallic cartridges for rifled arms have longer ranges than those of handguns. They are usually longer than pistol cartridges, contain more propellant, and are designed for longer barrels, which allow bullets to be fully accelerated. This provides more velocity and energy to the bullet, increasing its range and penetration power.



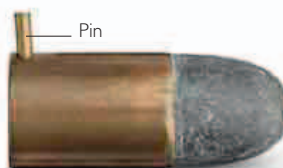
PAPER CARTRIDGES

The first cartridges were nothing more than paper packages containing a measured charge of powder and a ball. They were used with both flintlock and percussion systems.



WESTLEY RICHARDS “MONKEY TAIL” CARTRIDGE

This paper-wrapped cartridge had a greased felt wad at the rear, which remained in the breech until pushed forward for removal before a new round was loaded. Doing so cleaned the bore and reduced fouling.



PIN-FIRE CARTRIDGE

Invented in the 1830s, the pin-fire was an early version of the unitary metallic cartridge. When the trigger was pulled, the gun’s hammer fell on a pin projecting from the base of the cartridge. The force of impact drove the pin into the primer contained within the cartridge’s base, igniting the primer and firing the gun.



SNIDER-ENFIELD BOXER CARTRIDGE

This was an early experiment at producing a center-fire cartridge, in the 1860s, with the primer at the center of the base. This cartridge for the Snider-Enfield rifle had a perforated iron base and walls built up from coiled brass foils.



.56IN-50 SPENCER (1860)

The rim-fire was another early type of metallic cartridge. This rim-fire round was fired by the first effective repeater rifle—the Spencer carbine—from the Civil War-era.



11MM CHASSEPOT (1871)

After the Franco-Prussian War (1870–71), the cartridge developed for the Mauser M71 rifle was adapted for the Chassepot rifle, which was converted to take it.



.30IN-30 WINCHESTER (1895)

This cartridge was the first “civilian” round to be charged with smokeless powder (see pp.142–43), a new propellant. It contained 30 grains (1.94g) of it.



.303IN MK V (1899)

The British Army’s Lee-Metfords and Lee-Enfields were chambered for this blunt-nosed rifle bullet from 1899.